

BIOLOGY BOARD QUESTION PAPER: MARCH 2024**Time: 3 Hrs. Max. Marks: 70****General Instructions:**

The question paper is divided into four sections.

- (1) Section A: Q.1 contains Ten multiple choice type of questions carrying one mark each.
Evaluation will be done for the first attempt only. Q.2 contains Eight very short answer type questions carrying one mark each.
- (2) Section B: Q.3 to 14 are short answer type questions carrying two marks each.
(Attempt any Eight)
- (3) Section C: Q.15 to 26 are short answer type questions carrying three marks each.
(Attempt any Eight)
- (4) Section D: Q.27 to 31 are long answer type questions carrying four marks each.
(Attempt any Three)
- (5) Begin the answer of each section on a new page.

SECTION – A

Q.1 Select and write the correct answer for the following multiple choice type questions: [10]

i. Identify the growth hormone in plants which causes inhibitory effect.

(A) Cytokinins (B) Absciscic acid (C) Gibberellin (D) Ethylene

ii. Which one of the following is not a part of lac operon?

(A) Promoter (B) Regulator (C) Inducer (D) Operator

iii. In absence of fertilization, corpus luteum degenerates into _____.

(A) tunica albuginea (B) membrana granulosa (C) zona pellucida (D) corpus albicans

iv. Which of the following divides nasal cavity?

(A) Hyaline cartilage (B) Mesethmoid cartilage (C) Ligamentum arteriosum (D)
Laryngopharynx

v. Which of the following is caused by unsterilized needle?

(A) Elephantiasis (B) AIDS (C) Malaria (D) Dengue

vi. Recognition sequence of restriction enzymes are generally _____ nucleotides long.

(A) 2 to 4 (B) 4 to 8 (C) 8 to 10 (D) 14 to 18

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vii. Which of the following types require pollinator but result is genetically similar to autogamy?

(A) Geitonogamy (B) Xenogamy (C) Apogamy (D) Cleistogamy

viii. Which one of the following does not evolve further?

(A) Climax community (B) Primary succession (C) Pioneer species (D) Seral community

ix. Identify the appropriate term for the number of births under ideal conditions:

(A) Absolute mortality (B) Realized natality (C) Realized mortality (D) Absolute natality

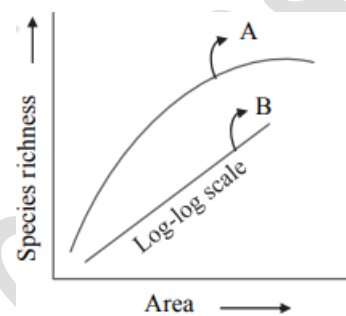
x. Observe the graph and select correct option:

(A) Line 'A' represent $S-CA^2$

(B) Line 'B' represents $\log C = \log A + Z \log S$

(C) Line 'A' represents $S = CAZ$

(D) Line 'B' represents $\log S = \log Z + C \log A$



Q.2 Answer the following questions: [8]

i. What are vestigial organs?

ii. Expand the term ZIFT.

iii. Name the endocrine gland prominent at birth but gradually atrophies in adult stage.

iv. What is the full form of IAA?

v. Give the name of microbial source of antibiotic chloromycetin.

vi. Which cells of islets of Langerhans produce insulin?

vii. How many meiotic divisions are required for formation of 300 seeds in angiosperms?

viii. Explain the term Emigration.

SECTION – B

(Attempt any EIGHT) [16]

Q.3 What are the reasons for the success of Mendel?

Q.4 Arrange the following steps of DNA fingerprinting in correct sequence:

i. Gel electrophoresis ii. Isolation of DNA iii. Southern blotting iv. Restriction digestion

Q.5 Distinguish between human sperm and ovum.

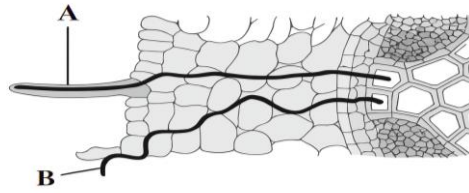
Q.6 Enlist the uses of gene therapy.

Q.7 Define the following terms: i. Gene flow ii. Chromosomal aberrations

Q.8 What are the significances of double fertilization?



Q.9 Identify and define 'A' and 'B' in relation to uptake of water by the root.



Q.10 Describe mutualism.

Q.11 Explain the factors affecting water absorption.

Q.12 What is differentiation and redifferentiation?

Q.13 Select and rewrite appropriate disorder of respiratory system:

(sinusitis, emphysema, silicosis and asbestosis, laryngitis)

i. Breakdown of alveoli, shortness of breath.

ii. Inflammation of the sinuses, mucous discharge.

iii. Inflammation of larynx, sore throat, hoarseness of voice, mucous build up.

Q.14 Explain the steps involved in preliminary treatment of sewage.

SECTION – C

(Attempt any EIGHT) [24]

Q.15 Give the steps involved in formation of m-RNA from hn-RNA.

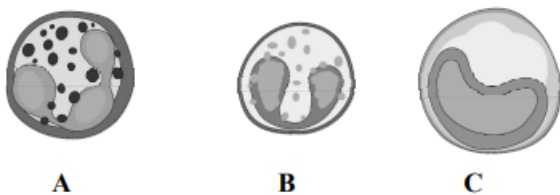
Q.16 What is reproductive isolation? Describe any two types each of pre-mating and post-mating isolating mechanisms.

Q.17 Explain unique features of acquired immunity.

Q.18 Name and describe hormones secreted by ovaries.

Q.19 Explain different steps involved in PCR technique.

Q.20 Identify A, B and C from the diagrams and give their functions.



Q.21 What are the limitations of root pressure theory?

Q.22 Explain greenhouse effect with reference to gases responsible and their sources.

Q.23 Describe physiological effects and applications of ethylene.

Q.24 Give the name and type of I, IV and VII cranial nerves.

Q.25 Describe pyramid of energy with the help of diagram.

Q.26 What is lac? Enlist economic importance of lac.



SECTION – D**(Attempt any THREE) [12]**

Q.27 Describe histological structure of testis with well labelled diagram.

Q.28 What are chromosomal disorders? Describe Turner's syndrome and Klinefelter's syndrome.

Q.29 Describe nervous system in Planaria with well labelled diagram.

Q.30 Explain the following terms: i. Grafting ii. Apomixis iii. Polyembryony iv. Parthenocarpy

Q.31 Interpret the diagrams A and B. Enlist changes occurring during inspiration and expiration.

